Education Project

Abstract

This project explores the relationships between socioeconomic and geographic factors and school average ACT scores and then develops regression models to predict these scores. EdGap data were combined with NCES school information and geographic classifications, including 7,227 high schools across 20 states. After data preparation and exploratory data analysis, ordinary least squares regression compared single input, quadratic, and multiple linear regression models. A reduced model using unemployment, college proportion, and free or reduced-price lunch eligibility was selected due to its simplicity and nearly identical fit to the full socioeconomic model. This model explained 62.8% of observed ACT variation, with RMSE of 1.530 and MAE of 1.145 points. Lunch eligibility had the largest absolute coefficient after predictor standardization. Adding school locale gave a statistically significant but small improvement, so it wasn't selected to use in the final model.

1. Introduction

Student academic performance varies across high schools in the United States. This raises the question of whether the socioeconomic and geographical factors of a school's location influence student outcomes. One metric for academic performance is the average ACT score at each school. The primary objective of this project is to explore the relationship between socioeconomic and geographical factors and ACT scores, and subsequently develop a regression model to predict the average ACT score of a school based on these factors.

This project combines the 2016 EdGap dataset with school information from the National Center for Education Statistics (NCES), along with geographical classifications. EdGap provides school identifiers, average ACT scores, and various socioeconomic indicators. These indicators include the unemployment rate, the proportion of adults with a college degree, the percentage of children children living in married-couple families, median household income, and the proportion of students eligible for the free or reduced-price lunch program. Data from the NCES provides school identifiers and locale classifications: City, Suburban, Town, and Rural. After merging these data sources, filtering for high schools with available ACT score data, and imputing missing values, the final dataset includes 7,227 schools across 20 states.

2. Theoretical Background

Linear regression predicts a numerical response using one or more predictors. Simple linear regression uses one predictor, while multiple linear regression uses more than one. Ordinary least squares estimates model coefficients by minimizing the residual sum of squares, where each residual is the observed response minus its fitted value. In multiple regression, a coefficient represents the predicted change in the response associated with a one-unit increase in that predictor, holding the other predictors constant (James et al., 2023, pp. 71–72, 80–82). However, the true relationship between the response and the predictors may be non-linear. In this case, a quadratic

regression is used to measure a curved relationship while retaining a model that is linear in its coefficients (James et al., 2023, pp. 98–99).

Categorical predictor variables can be incorporated into a linear regression model through indicator variables that take on values of 0 or 1. When the model includes an intercept, a predictor with four categories is represented by three indicator variables, with one category serving as the reference group. In this project, "city" is selected as the reference category for the school location variable. Each coefficient corresponding to the other locations represents the difference in predicted ACT scores relative to the "city"group, assuming all other predictor variables remain constant (Bradbury & Eubank, n.d.).

To assess the accuracy of the model, there are some quantities used in this project. R^2 indicates how many the variation in the response variable is explained by the model. RMSE (Root Mean Squared Error) measures prediction errors, but penalizes larger errors more heavily. MAE (Mean Absolute Error) measures the average absolute difference between predicted and actual values. A higher R^2 generally indicates a better fit, while lower RMSE and MAE indicate smaller prediction errors (James et al., 2023, pp. 78–80, 238). Coefficient p-values assess evidence against a zero coefficient, while partial F-tests assess the joint contribution of additional terms in nested models under classical OLS assumptions (James et al., 2023, pp. 84–85). Residual plots complement these measures by revealing curvature, variations in error spread, and unusually large errors (James et al., 2023, pp. 100–101).

3. Methodology

The analysis used the data science methodology and was conducted in Python using pandas, NumPy, Matplotlib, Seaborn, scikit-learn, and statsmodels. Here are three main steps included:

Step 1 - Data preparation: The EdGap data was left-joined with NCES school information and geographic data using school identifiers. Locale codes were grouped into City (11–13), Suburb (21–23), Town (31–33), and Rural (41–43) categories. The analysis was restricted to high schools. Values below 0 for lunch eligibility and ACT scores below 1 were treated as missing, and schools with no ACT results were excluded. Missing values for the five numerical socioeconomic predictor indicators were imputed using scikit-learn's IterativeImputer. The clean dataset comprises 7,227 schools across 20 states.

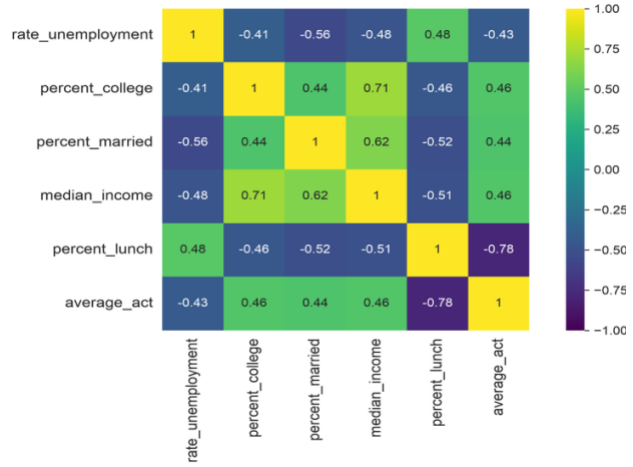
Step 2 - Exploratory data analysis: Descriptive statistics, histograms, and boxplots summarized ACT scores and socioeconomic characteristics. Potential outliers were identified using the 1.5 × IQR rule. Pearson correlations and pairplots examined numerical relationships. Grouped statistics and boxplots compared ACT scores and socioeconomic characteristics across locales.

Step 3 - Model development and assessment: There are six models. Lunch eligibility was selected for the single input model because it had the strongest absolute Pearson correlation with ACT among the socioeconomic variables. A quadratic model added its squared term. The full socioeconomic model included unemployment, college proportion, married-couple family

proportion, household income, and lunch eligibility. A reduced model included only unemployment, college proportion, and lunch eligibility. Both full and reduced socioeconomic models were then extended with three locale indicators, using City as the reference category. After model development, all models were compared using R^2, adjusted R^2, RMSE, and MAE. Partial F-tests were compared at a significance level of 0.05. Model selection considered both fit and simplicity. The selected model's numerical predictors were standardized using StandardScaler to support coefficient interpretation.

4. Computational Results

The analytical sample contained 7,227 schools across 20 states. School average ACT scores had a mean of 20.30, a median of 20.50, and a standard deviation of 2.51. The middle 50% of scores lay between 18.80 and 22.00, giving an interquartile range of 3.20 points.



From figure 1, we can see that lunch eligibility had the strongest negative correlation with ACT scores ($r = -0.784$). College proportion, household income, and married-couple family proportion were positively correlated with ACT scores, while unemployment was negatively correlated. Correlations were also observed among the socioeconomic predictors.

Figure 1. Correlation Matrix

Mean ACT scores were 18.91 for City, 21.08 for Suburban, 20.34 for Town, and 20.50 for Rural schools. City schools had the largest standard deviation (3.24), followed by Suburban (2.49), Rural (1.72), and Town (1.69). Figure 2 displays the ACT score distributions across school locales.

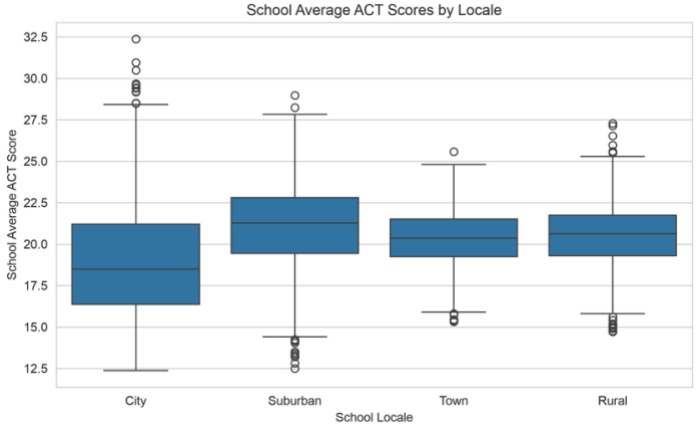


Figure 2. ACT score by locale

	R-squared	Adjusted R-squared	RMSE	MAE
Lunch only	0.6139	0.6139	1.5585	1.1690
Lunch quadratic	0.6158	0.6157	1.5548	1.1651
Socioeconomic full	0.6280	0.6277	1.5299	1.1453
Socioeconomic reduced	0.6279	0.6278	1.5300	1.1455
Full + locale	0.6285	0.6281	1.5289	1.1443
Reduced + locale	0.6285	0.6282	1.5289	1.1443

Table 1. Model Comparison

Table 1 shows that the lunch-only model achieved an R^2 of 0.6139 and an MAE of 1.1690 ACT points. The full and reduced socioeconomic models had nearly identical results, with R^2 values of 0.6280 and 0.6279 and MAE values of 1.1453 and 1.1455, respectively. Adding locale reduced MAE to 1.1443 for both specifications.

The reduced socioeconomic model is selected for its simplicity and nearly identical in-sample fit to the full model. Although adding locale produces a statistically significant improvement, the reductions in RMSE and MAE are very small. The selected model includes unemployment rate, college proportion, and lunch eligibility as below:

Predicted ACT = 22.6400 − 2.1683U + 1.7139C − 7.5863L, where U, C, and L denote unemployment, college proportion, and lunch eligibility, respectively, on a 0-1 scale. All three predictor coefficients had $p < 0.05$. After standardizing the predictors, their coefficients were −0.123, 0.283, and −1.777, respectively.

5. Discussion

Exploratory data analysis (EDA) showed that schools with a higher proportion of students eligible for free or reduced-price lunches tended to have lower average ACT scores. A model based only on lunch explained 61.4% of the variance, compared to 62.8% for a full socioeconomic model. In the reduced model, after standardizing the predictor variables, lunch eligibility was the variable with the largest absolute coefficient. Therefore, these findings indicate that this variable provided useful information for predicting ACT scores, but it doesn't mean that receiving a free lunch is the cause of lower academic performance.

Geographic comparisons showed that suburban schools had higher average ACT scores than urban schools, in addition to higher household income and lower lunch eligibility. Adding locale to the reduced socioeconomic model decreased MAE by only 0.0012 ACT points. This means some of the differences observed across locales overlap with socioeconomic differences.

The reduced socioeconomic model was selected because it used fewer predictors while retaining almost the same fit as the full socioeconomic model. Although household income and married-couple family proportion were positively correlated with ACT in the EDA, they added little information once the other predictors were included. Similarly, the statistically significant quadratic lunch term reduced MAE by only 0.0039 points. These comparisons support choosing a simpler model.

6. Conclusions

A regression model using unemployment, college proportion, and lunch eligibility explains approximately 62.8% of the observed variation in school average ACT scores, with an in-sample RMSE of 1.530 and MAE of 1.145 ACT points. It is nearly the same fit as the five-predictor socioeconomic model, while adding school locale provides only a small further improvement. Therefore, the reduced model was selected for its simplicity. These findings show that socioeconomic characteristics, particularly lunch eligibility, provide useful information about differences in academic performance across schools.

However, the analysis is limited to 7,227 schools across 20 states. Large prediction errors remain for some schools, and all performance metrics are calculated based on the available data. Furthermore, findings at the school level cannot be directly applied to individual students.

7. References

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